Indian Institute of Technology Bhilai
Department of Computer Science Engineering

Network Science (CSL601/CS552)

Mid Sem, Date: Feb 19, 2024

Timing: 11:30 AM to 1:30 PM Max mark: 60

1. Mark the correct option. Mark multiple options if multiple options are correct. **(6)**
   (a) What is the programming language used by GraphViz?
   (a)Python (b)DOT (c)C (d).NET
   (b) What are the export options available in GraphViz?
   (a)PNG (b)PDF (c)JPEG (d)PostScript
   (c) What are the different formats available to load networks in SocNetV?
   (a)GrpahML (b)GML (c)UCINET (d)EdgeList
   (d) Different network visualizations available in SocNetV.
   (a)Prominence based (b)Force-Directed (c)Yifan Hu (d)Force Atlas
   (e) How do we calculate eigenvector centrality in Pajek?
   (a)Hubs-Authorities (b)Proximity Prestige (c)Laplace (d)Centers
   (f) Can edges be assigned weights as strings in an edge-weighted graph in Networkx?
   (a)Yes (b)No

2. Consider the following network represented by the graph: **(14)**

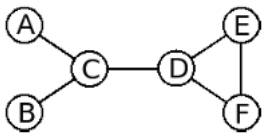


Figure 1:

(a) Calculate each node's (normalized) betweenness and closeness centrality.

(b) Calculate the degree centrality of each node.

(c) Calculate the local clustering coefficient of each node in the network.

3. Find the eigenvector centrality for the following network. **(10)**

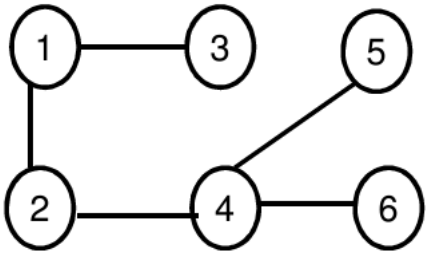


Figure 2:

4. (a) Start with a ring lattice with each node being connected to one neighbor on each side(that's one neighbor to the left, and one neighbor to the right). Sketch the network. What is the average clustering coefficient for this network? **(5)**

(b) Assume that the number of nodes in the ring lattice is even, and that the nodes are numbered from 1 to n. In addition to the edges specified in (part (a)), connect the closest even nodes to

one another(e.g. Node 4 connects to nodes 2 and 6, and node 2 connects to node n and node 4). Sketch the network. What is the average clustering coefficient? **(5)**

5. Consider the following network represented by the graph to find: **(10)**

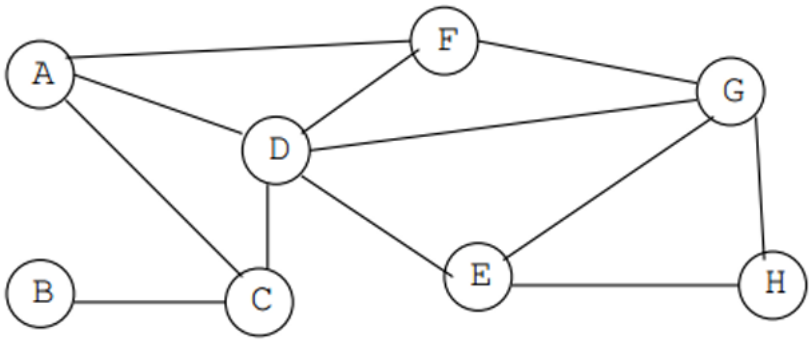


Figure 3:

(a) Graph Density

(b) Diameter

(c) Average Path Length

(d) Average Degree

(e) Longest Path

(f) Show the degree of each node and make a plot of its degree distribution.

6. Consider a network generated using Erdos-Renyi Graph Model $G(n, p)$. **(10)**

(a) What is the expected number of edges in terms of n and p?

(b) Explain the significance of the parameter p in the Erdos-Renyi random graph model.

(c) What happens as p approaches 0 and as p approaches 1?

(d) How does the expected degree of a vertex change as n increases, assuming p remains constant?

(e) Under what conditions can we expect a connected component containing a significant fraction of the vertices to emerge?